

Assembling the pieces of the reference resolution puzzle

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1 Introduction

- Considering current theory and the delayed acquisition of polite pronominal and reflexive anaphora, I argue that the third-factor operation Merge ((Hornstein, 2001, 2013; Richards, 2020), constrained by phases, rather than ϕ or non- ϕ Agree (Reuland, 2011; Rooryck & Wyngaerd, 2011; Adger & Ramchand, 2005; Hicks, 2009; Sundaresan, 2012, 2020), is the optimal candidate for reference resolution. I make three sub-arguments:
 - Feature and feature values involve a high degree of variation across languages. Further, anaphor-antecedent agreement is perhaps not the same beast as controller-target agreement. Therefore, appealing to Agree to resolve anaphora burdens the acquisition path, a burden for which children do not show evidence.
 - Anaphora resolution unfolds at the vP phase mediated by two features, [REF(ERENTIAL)] (morphosemantic) and [REFL(EXIVE)] (morphosyntactic), and is subject to phasal restrictions.
 - The legibility conditions on FL imposed by the conceptual-intentional (C-I) interface appeals to Merge, and not Agree. I posit a late insertion rule to produce the right morphological forms of the pronouns and reflexives. This meets the demands of the SM interface.

2 Problems with Agree-based reference resolution

- The traditional account of anaphora resolution assumes that anaphors are ϕ -deficient (Kratzer, 2009; Reuland, 2011; Rooryck & Wyngaerd, 2011).
 - Features already present in the inventory of the language participate in anaphora resolution.
 - This recycling of features makes this theoretically economical and thus appealing.
 - There is also the fact that anaphors match their antecedent in ϕ -features. This is morphologically visible.
- The less trodden path is one where anaphors are argued to be deficient in non- ϕ features. These could be the something like the dependency feature ID (Adger & Ramchand, 2005), the referential feature VAR (Hicks, 2009), or the perspectival feature DEP (Sundaresan, 2012, 2020).

2.1 Piggybacking on Case

- Downward Agree-based explanations of reference resolution resort to having anaphor-antecedent agreement piggyback on Case licensing (Reuland, 2011).
- Why should reference assignment, which is ultimately an assignment of values to nominals based on assumptions about a given world, therefore a semantic assignment, be dependent on a purely syntactic/morphosyntactic phenomenon? Letting anaphora resolution piggyback on Case licensing seems ad hoc.

2.2 Features display high degree of variation

- There are features whose values remain constant across contexts — Number, Noun Class.
- There are features whose values vary with context — natural gender, politeness, even person.
 - Malayalam allows for proper nouns and titles, which are usually considered third-person renditions, to be used to refer to the second person. Therefore, a sentence like (1) is ambiguous. *Iqbal/Sir* may refer to a third person or the addressee (second person).

(1) *Malayalam*

ikbaal/saar entha innale varaathe
Iqbal/sir what yesterday come.PST.NEG

‘Why didn’t Iqbal/Sir come yesterday?’ (‘Iqbal/Sir’ can either refer to the addressee or a third person.)

- Malayalam also allows for the first person inclusive plural to refer to the second person. This is a strategy of expressing solidarity with the addressee.
- Features are also treated differently in typology, morphosyntax, and semantics.

2.3 Features with no morphological manifestation

- The primary issues with features like the ID feature and VAR are that they are posited only for building a dependency, i.e., they do not show up as operational elsewhere in the language, and that they are syntactic features with only a logical/formal semantic use, i.e., there is no morphosyntactic evidence to support their existence.
- While the first issue holds for Sundaresan (2020)’s perspectival feature DEP as well, the second does not. In languages like Tamil, Icelandic, and Italian, the anaphor must not only match the antecedent in ϕ -features but is also argued to contain information about the perspective holder (Jayaseelan, 1998/2017).
- The Tamil anaphor *Taan* is posited to have a set of Person features along with an unvalued DEP feature. It is the valuation of DEP that translates into semantic binding (Sundaresan, 2020, p. 436). Like the VAR feature, the DEP feature is “an attribute-value pair that takes arbitrarily assigned integers/letters as value” (p. 436). Again, this means interpretability and inclusiveness condition violations.

2.4 Anaphora agreement may not be controller target agreement

- The anaphor and its antecedent match in featural values.
- Which features are involved in the link between the anaphor and its antecedent? Are these what we know to be the traditional morphosyntactic-/ ϕ -features, or are these other kinds of features, or a combination of both?
 - Is the operation underlying anaphor-antecedent feature identity the same as that underlying controller-target feature identity?
- Similarities between the two (first three also noted by Murugesan (2019)).
 1. Both involve ϕ -feature matching/dependency between a controller/antecedent and a target/anaphor
 2. Both require that the controller/antecedent c-command the target/anaphor, and
 3. This relation breaks down when there is an intervening entity i.e. when some X c-commands the target/anaphor and is c-commanded by the controller/antecedent.
 4. Both are sensitive to locality and yet both are known to exhibit long-distance properties i.e. there exists both long-distance agreement and long-distance binding.
- Examining the similarities.

- 1 is descriptively accurate, yet phi-features on the target are uninterpretable, while they are interpretable on the anaphor
- C-command is a consequence of the derivational process of structure building via Merge (Chomsky, 1995; Epstein, 1999; Reuland, 1998). All dependency relations will essentially use this (language) universal mechanism. Intervention effects arise from the requirement that all dependency relations be local, at least at the point where they are interpreted. Again, this is a universal property of all syntactic dependency relations. Both 2 and 3 can be argued to be requirements of LF since LF translates the configuration provided by the syntax into a binding relation.

• Differences between the two:

1. **Agreement blocking:** Nominal-verb agreement may be blocked if the nominal has morphological case marking.

The subject is in the nominative in the Bangla utterances in (2a) and (2b). The Bangla verb agrees in person and politeness with the subject. However, when the subject is dative or genitive case marked, as in (3a) and (3b), ϕ -feature agreement on the verb collapses to the default, which is third person, L2 politeness.

- (2) a. apnara ama-r bari age eʃetʃ^h-en
2.PL.L3.NOM 2SG-GEN home before come.PRF.PRS-3L3
'You (all) have come to my house before.'
- b. ami toma-r bari age eʃetʃ^h-i
1SG.NOM 2.SG-GEN house before come.PRF.PRS-1
'I have come to your house before.'
- (3) a. ona-ke onek dāt dek^hte hol-o
3.SG.L3-DAT many teeth see.NF be.PFV-3L2
'She/he had to see many teeth.'
- b. ona-r k^hub k^hide pejetʃ^h-e
3.SG.L3-GEN very hunger get.PRF.PRS-3L2
'She/he is very hungry.'

A case marker on a nominal does not block the nominal from being a potential antecedent of an anaphor.

2. **Manifestation in the lexicon:** While anaphors can be argued to be stored in the lexicon as it is, the entities that undergo agreement in the verbal and nominal domains, i.e., verbs and noun modifiers, cannot be said to be stored as it is [*himself*, *herself* and *themselves* vs *is*, *was*, and *were*]
The features that would compose the reflexives, for example, can be argued to be features that these items are born with in the lexicon, i.e., they would be of semantic relevance; targets of agreement cannot be argued to be born with the features that ultimately show up on them.
3. **Interpretability of features:** ϕ -features on pronouns and reflexives are interpreted and are therefore crucial for the C-I system, agreement features of the subject that are copied onto the verb, or those of the head nominal that are copied onto the modifiers, are not.
4. **Distribution:** Anaphor-antecedent agreement is attested in all languages with anaphors. The anaphor usually shows some morphosyntactic agreement or the other with the antecedent. However, some languages have no controller-target agreement. E.g., Malayalam — Its anaphor *taan* agrees in number with the antecedent. Further, it only appears when the antecedent is in the third person. However, Malayalam does not attest to either noun-verb agreement or agreement/concord in the nominal domain between modifiers and nouns.

3 Acquisition of some features is delayed

- Evidence from the acquisition of polite pronouns shows that there is a steep learning curve in the acquisition of polite, including features that build polite pronouns in Bangla (Roy, 2023).
- Children aged <6;11 replaced formal pronouns with non-formal ones, producing infelicitous utterances.

(4) *Pictorial context: A little girl tickling her mother.*

BCET1, bch24: 2;8

RE: *ʃʊʃʊɾi dɪtʃtʃʰe dækʰo to*
tickles give.PROG.PRS.3.L2 see.IMP.2.L2 TO

Is she tickling (someone)? Have a look!

CH: *hǎ*
yes

Yes

CH: *hǎ ʃʊʃʊɾi dɪtʃtʃʰe*
yes tickles give.PROG.PRS.3.L2

Yes, she is tickling (someone)

RE: *kake dɪtʃtʃʰe ʃʊʃʊɾi*
whom give.PROG.PRS.3.L2 tickles

Whom is she tickling?

CH: **o-ke dɪtʃtʃʰe**
3SG.L2-DAT give.PROG.PRS.3.L2

She is giving her (tickles)

(OI: pointer on woman in BCP1-3)

CH: **o-ke o-ke**
3SG.L2-DAT 3SG.L2-DAT

Her! Her!

- In contrast, adults, while also underusing formal pronouns, perhaps owing to the characters not necessarily being older than the adults in the tasks, make felicitous replacements. Adults replaced the formal pronoun felicitously with bare nouns, common nouns, and pseudo-kinship terms.

(5) Common Noun replacement by adult (bad02, BAET3)

pensil dije natni-ke boktʃʰe
pencil with granddaughter-ACC scold.PROG.PRS.3.L2

‘She_i is scolding her_i granddaughter with a (/her_i) pencil.’

- No child produced the Bangla *je*-set pronouns, which are known to be used to signal politeness.

(6) *je* used to signal politeness

a. *profesar daʃ adʒ klas nebe-na #o confarens-er priparaʃon nije*
Professor Das today class take.INDF.PRS.3.L2-NEG 3SG.L2.NOM conference-GEN preparation with
bæsto
busy

Intended: ‘Professor Das will not take class today. #She (L2) is busy preparing for the conference.’

b. *profesar daʃ adʒ klas nebe-na tini confarens-er priparaʃon nije*
Professor Das today class take.INDF.PRS.3.L2-NEG 3SG.F.NOM conference-GEN preparation with
bæsto
busy

‘Professor Das will not take class today. She (F) is busy preparing for the conference.’

- One child produced utterances where they replaced the personal pronoun with demonstrative ones (5 instances). The language allows for demonstrative pronouns to be used to refer to human referents when the speaker wishes to convey disrespect or disdain towards the referent.

(7) *Pictorial context: A little girl tickling her brother*

BCET1, bch21: 3;6

RE: *æk-ta tʃʰele*
one-CLF boy

One boy

(Pic: BCP3-1)

RE: amar mone hotʃʰe eta or bon
 1SG.NOM mind.LOC be.PROG.PRS.L3 this -3SG.L2-GEN sister
 I think this is his sister (Pic: BCP3-2)

RE: or bon ki kortʃʰe
 3SG.L2-GEN sister what do.PROG.PRS.L2
 What is his sister doing? (Pic: BCP3-3)

CH: hāstʃʰe
 smile.PROG.PRS.L3
 She is smiling

RE: ʃurʃruʃi ditʃʰe
 tickle give.PROG.PRS.L3
 Tickling?

RE: kake
 whom
 Whom?

CH: eta-ke
 this-ACC
 This one

4 Anaphora resolution in the vP domain, sensitive to phase-hood

- Reflexivity is a property of the predicate [also argued for by Reinhart and Reuland (1993)]. Reflexivisation is licensed in the syntax in the domain where the predicate and argument structure is built, i.e., in the vP domain. It is driven by the interpretable morphosyntactic feature [REFL] on the v-head. The C-layer/clausal domain plays no role in licensing reflexivity.
- Four pieces of evidence: The first three are about which nominals are permitted as linguistic antecedents of reflexives and bound pronouns when the predicate is altered. The fourth piece of evidence is from the structure of Malayalam complex reflexives.

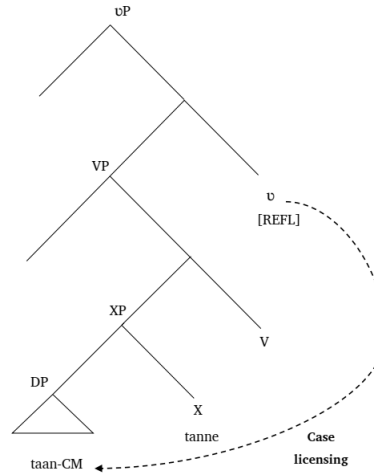
- **Antecedence in embeddings with ditransitives:** When a ditransitive is embedded, and the reflexive/pronoun is in(side) the indirect object position, the ditransitive in question determines whether the nearest subject (embedded clause subject) and/or the next higher subject (main clause subject) can be a potential antecedent.

- (8) a. ʃatʰ_i dʒane [CP dʒe ʃomir_j dʒonmodin-e nidʒe_{*i/j}-ke boi dijetʃʰe]
 Sathi know.PRS.INDF.3.L2 C Samir birthday-LOC REFL-DAT book give.PRS.PRF.3L2
 ‘Sathi_i knows that Samir_j gave himself_{*i/j} a book for his birthday.’
- b. ʃatʰ_i dʒane [CP dʒe ʃomir_j nidʒ_{*i/#j}-er katʃʰ tʰeke ʈupi-ʈa
 Sathi know.PRS.INDF.3.L2 C Samir REFL-GEN near from hat-CLF return
 pʰerot nebe]
 take.FUT.3L2
 ‘Sathi_i knows that Samir_j will take the hat from self_{*i/#j}.’

- **Antecedence in embeddings with possessives:** The second piece of evidence is from which higher nominal is permitted as a potential antecedent of a possessive reflexive in the object of the embedded clause when the main clause verb is simple as opposed to complex.

- (9) These examples are inspired from Haque (2003, p.15), example (16b)
- a. ami_i mala-ke_j [PRO_{*i/j/*k} nidʒe_{*i/j/*k}-r upor eksperiment korte] bollam
 1.SG.NOM Mala-DAT [PRO REFL-GEN on experiment do.NF] say.PFV.1
 ‘I_i told Mala_j to do experiments on self_{*i/j/*k}.’
- b. ami_i mala_j-ke [PRO nidʒe_{i/j/*k}-r upor eksperiment korte] dilam
 1.SG.NOM Mala-DAT [PRO REFL-GEN on experiment do.NF] give.PFV.1
 ‘I_i allowed [Mala_j to do experiments on self_{i/j/*k}].’

Figure 1: Structure of *taan-e tanne*



- **Bare body part direct objects:** The third piece of evidence is from clauses with bare body parts as direct objects. Bare body-part or kinship address term nominals in object position in Bangla can have a possessive reading, with the nearest subject as their antecedent. Consider the sentence in (10).

- (10) mala pa d^hutʃtʰe
 Mala feet wash.PROG.PRS.3.L2
 ‘Mala_i is washing her_i feet.’

This is the case when the verb is simplex. Some complex verbs cause the loss of this meaning, i.e., whether *pa* can refer to the feet of someone who is not the referent of the subject is dictated by the predicate.

When the predicate encodes information that the agent and the patient are distinct, i.e., the agent benefiting a patient, the bare body part is that of the patient and not the agent. Therefore, in (11a), the feet belong to that of the addressee and not the speaker. And in (11b), the feet belong to *Madhobi*.

- (11) a. toma-r pa-ʃa k^hub noŋʃa hoj gætʃ^he tʃ^holo ami pa-ʃa d^huje
 2SG.GEN feet-CLF very dirty be.NF go.PROF.PRS.3.L2 come 1SG.NOM feet-CLF wash.NF
 di
 give.SUBJ.1
 ‘Your feet are very dirty. Come, let me wash your feet.’
 b. mad^hodi-r pa k^hub noŋʃa tai dʒonne ʃila pa d^huje ditʃtʰe
 Madhobi-GEN feet very dirty. that-I for Sheela feet wash.NF give.PROG.PRS.3.L2
 ‘Madhobi’s feet are very dirty. That is why Sheela is washing the feet.’ (The feet must belong to *Madhobi*, and not to someone else, particularly not Sheela’s, otherwise, the second utterance will be infelicitous.)

- **The structure of the Malayalam complex reflexive:** Object position reflexives in Malayalam are complex in morphology. They have the structure *taan*-CM (PP) *TANNE*.

The casemarker linearly intervenes between *taan* and the post-position, or *tanne*. Case licensing does not happen in the lexicon. One can argue for it being operational in the syntax. Essentially, the position of the case marker hints towards the reflexive being ‘built’ in the syntax. When *v* contains information that the predicate is to be reflexivised, a reflexive nominal may be built.

- There is evidence from Bangla for the morphological existence of a feature that codes whether a nominal is referential or not. The [REF] feature has a semantic distribution and no specific syntactic distribution.
 - Bangla has two sets of pronouns — the *o*-set and the *ʃe*-set, as I christen them.
 - The *o*-set allows only permits referential antecedents. The *ʃe*-set allows for antecedents that are referential, quantificational, and even entire propositions.

Figure 2: Structure of *taan-e patti tanne*

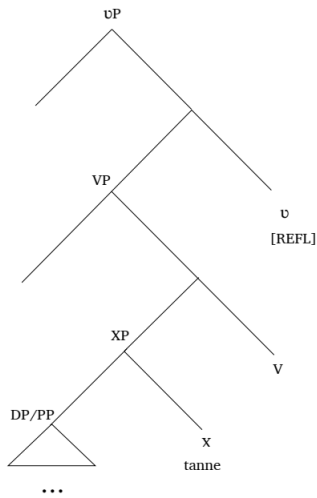
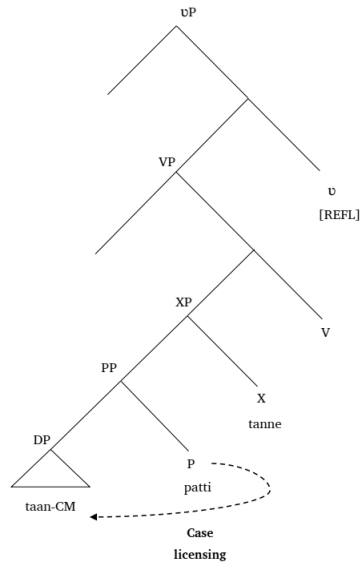


Figure 3: The REFL feature reflexivises the predicate

(12) Antecedent referential

- a. [tʰele-gulo]_i o_i-de-r bəl-ʈa harije pʰeletʰe
 boy-CLF 3.REF.L2-PL-GEN ball-CLF lose.NF throw-PRF-PRS.3.L2
 ‘[The boys]_i have lost their_i ball.’
- b. [tʰele-gulo]_i ta_i-de-r bəl-ʈa harije pʰeletʰe
 boy-CLF 3.REF.L2-PL-GEN ball-CLF lose.NF throw-PRF-PRS.3.L2
 ‘[The boys]_i have lost their_i ball.’

(13) a. Antecedent Non-referential

- keu_i ta_i-r pen tʰeɾe gatʰe
 someone 3SG.NREF.L2-GEN pen leave.NF go-PRF-PRS.3.L2
 ‘Someone (non-specific) has left her/his pen.’
- b. *keu_i o_i-r pen tʰeɾe gatʰe
 someone 3SG.REF.L2-GEN pen leave.NF go-PRF-PRS.3.L2
 Intended ‘Someone (non-specific) has left her/his pen.’

- The fact that reflexives and pronouns are not always in complementary distribution, contrary to the direct inference from Principles A and B of the Canonical Binding Theory (CBT), was noted for the possessor position of a nominal in Huang (1982) [see (14) from Mandarin Chinese]. The possessive reflexive and the possessive pronouns in Bangla too manifest in non-complementary distribution [see (15)]. This is a challenge for the CBT.

(14) Huang (1982, p. 325), example (20)

- a. *Chinese*
 zhangsan_i kanjian-le [ziji_i de shu]
 Zhangsan see-ASP self DE book
 ‘Zhangsan saw his own book.’
- b. Zhangsan_i kanjian-le [ta_i de shu]
 Zhangsan see-ASP he DE book
 ‘Zhangsan saw his books.’

(15) *Bangla*

- a. mala_i o_i-r boi potʰe
 Mala 3SG-GEN book read.PROG.PRS.3L2
 ‘mala_i is reading her_i book.’
- b. mala_i nidʒ_i-er boi potʰe
 Mala REFL-GEN book read.PROG.PRS.3L2
 ‘mala_i is reading self_i’s book.’

- This non-complementary distribution is a direct fallout if one were to describe binding domains in terms of phases and phasal properties, rather than phrases
- The domain in which an object position reflexive should find its antecedent is always a phase (the vP phase) rather than a phrase (Contra CBT). Phases are better candidates for potential binding domains because their formation in the syntax and their spellout are naturally constrained by locality conditions.
- Resorting to reducing binding domains to phases can explain the non-complementarity of possessive reflexives and pronouns in languages like Bangla, Malayalam if we were also to assume the following: (1) Bangla and Malayalam have DPs or DP-equivalent structure (2) The DP is a phase in these languages (See Citko (2014); Željko Bošković (2005, 2014); Matushansky (2005) for phasehood tests.)
- For the reflexive and the pronoun in the DP’s specifier, the DP phase is a local domain. If the possessive reflexive or pronoun is in the specifier of the DP and if the DP is a phase in Bangla, an immediately higher v will be able to access the specifier of DP. This will allow the reflexive or the pronoun to be bound by an operator that scopes over the subject DP (in the specifier of vP), and the possessive DP.

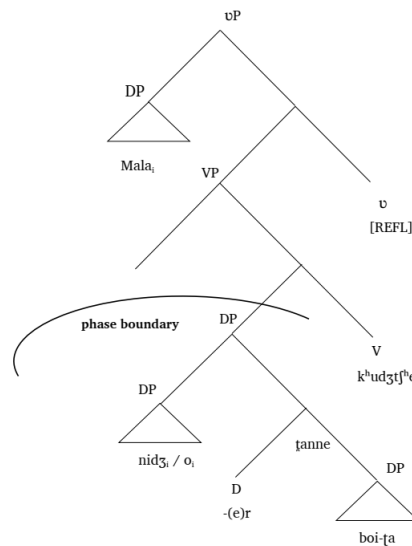


Figure 4: The licensing of the possessive reflexive or the pronoun

- The resolution of the possessive reflexive anaphora, although technically violating Principle A, does not cause derivational crash, and the resolution of the possessive bound pronoun adheres to Principle B.
- The reflexive in the possessive position of a DP is at its very edge, thus any potential antecedent would be outside this DP and higher up. Indeed, the reflexive is not bound in this local domain. This is a Principle A violation. However, being at the edge of the phase, it is visible to the next higher phasal head, which mediates binding between the subject of the vP and this possessive reflexive. Thus, Principle A can be redefined as follows.

- (16) Principle A: A reflexive must be bound within the nearest phase unless at the edge of the phase, in which case it can be bound within the next higher phase.

The possessive pronoun is free in the DP phase in which it is contained, thus following Principle B. Nothing stops it from being bound outside this domain, which is how it being bound by an operator scoping over it and the subject of the vP causes the derivation not to crash.

5 Merge and Late insertion to resolve anaphora

- Merge is innate/hardwired.
 - Not an operation exclusively for building language structure. It is available from the laws of nature. Think snowflakes, broccoli, and the successor function.
 - There is evidence from studies on the acquisition of c-command and binding to suggest this (Crain & Nakayama, 1987; Flynn & Lust, 1980; O'Grady, 1997; Musolino, Crain, & Thornton, 2000; Lidz & Musolino, 2002; Grolla, 2012; Chien & Wexler, 1990).
- Being the simplest operations and attesting early knowledge in children makes it the right candidate for driving the syntax of reference resolution.

5.1 Match, rather than Agree

- If anaphor-antecedent agreement is different from controller target agreement, the operation underlying these two agreements may not be the same.

- The way we talk about anaphor-antecedent agreement is to say something along the lines of (17). This presupposes that anaphors are deficient in feature values and that this valuation happens via Agree or an Agree-like operation. Thus, in this formulation, reflexives and pronouns can be posited as (highly) underspecified [see (Kratzer, 2009) for an account of underspecified pronouns].

(17) The anaphor agrees with its antecedent in feature values.

- If I paraphrase (17) as (18), the requirement for Agree disappears. ‘Match’ here is essentially feature identity roughly in the sense of Chomsky (2000). Match, unlike Agree would require that reflexives and pronouns be fully specified for features.

(18) Anaphors and their antecedents match in or have the same features.

About Match Chomsky says the following.¹

matching is a relation that holds of a probe P and a goal G. Not every matching pair induces Agree. To do so, G must (at least) be in the domain D(P) of P and satisfy locality conditions. (p. 122)

- Appealing to Match instead of Agree is also, as we will see, better for a ‘just Merge’ based account of reference resolution.

5.2 The legibility conditions & an implementation

- Assume the Strong Minimalist Thesis.
- Copies are formed by internal Merge and repetitions by external Merge.
- Linguistic items that are neither copies nor repetitions are formed by external Merge. The C-I interface needs to distinguish between copies and repetitions. This is how it knows, for example, to distinguish between ‘his’ in ‘he₁ likes his₁ ball’ from the ‘his’ in ‘he₁ likes his₂ ball’. If it couldn’t distinguish between the two, it would not know to resolve the former *his* via binding and the latter via covaluation.
- I posit examples (19), (20) and (21) below as illustrations of LC1, LC2 and LC3 respectively. This part of the proposal is inspired by Richards (2020), with the machinery of copies and repetition from Chomsky (2019).

5.2.1 LC-1 — leads to the creation of copies

C-I demands that the two linguistic items be interpreted as referring to the same individual/entity in the given world.

SM demands that the two linguistic items look distinct or identical.

(19) Copies (reflexive/bound readings)

a. Identity at C-I, distinction at SM

i. raaman₁ tann₁-e patti tanne paraŋnu₁
Raman REFL-ACC about TANNE say.PFV
‘Raman₁ talked about himself₁.’

ii. mala₁ [nidʒe₁-r vailin-[a] b^heɲe p^heletʃ^he
Mala REFL-GEN violin-CLF break.NF drop.PRF.PRS
‘Mala₁ has broken [self_{s1} violin₁].’

iii. mala₁ [o₁-r vailin₁-[a] b^heɲe p^heletʃ^he
Mala 3SGL2-GEN violin-CLF break.NF drop.PRF.PRS
‘Mala₁ has broken [her₁ violin].’

b. Identity at C-I, identity at SM

Hornstein (2001), Example (2c.)

Hmong

pov₁ yeej qhuas pov₁
Pao always praise Pao

‘Pao₁ always praises himself₁.’

Figure 5: Derivation of *raaman tann-e patti tanne paraynu*

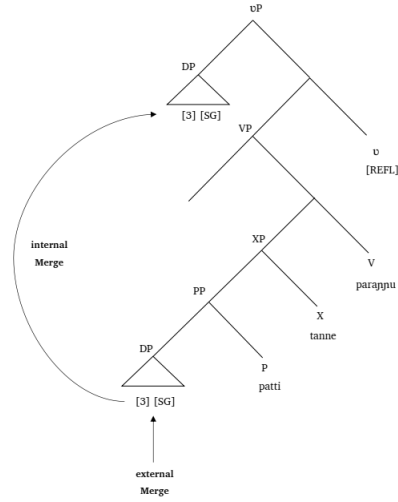


Figure 6: Derivation of *mala nidze-r vailin-ta b^heṇe p^helet^he*

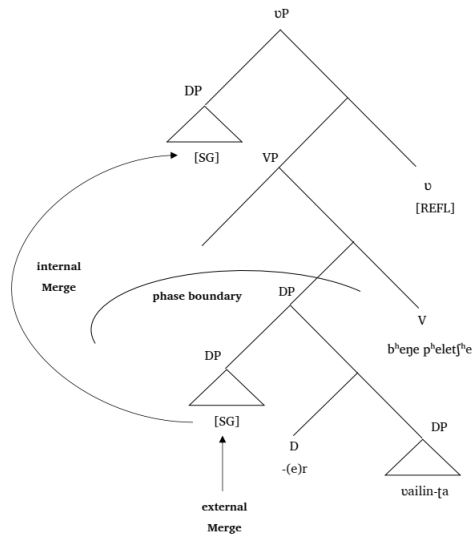
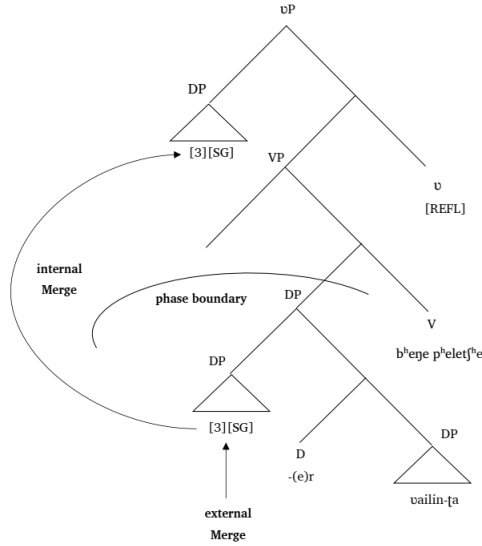


Figure 7: Derivation of *mala or vailin-ta b^heje p^helet^he*, where or is bound



5.2.2 LC-2 — leads to the creation of repetitions

C-I demands that the two linguistic items be interpreted as referring to distinct individuals/entities or the same individual/entity in the given world.

SM demands that the two linguistic items look distinct or identical.

(20) Repetitions (Covaluation)

a. Identity at C-I, identity at SM

mala mala-r vailin-ta b^heje p^helet^he
Mala Mala-GEN violin-CLF break.NF drop.PRF.PRS

‘Mala₁ has broken [Mala₁’s violin].’

b. Identity at C-I, distinction at SM

mala bollo dze jon o-ke t^hene
Mala say.PFV C John 3SGL2-ACC know.INDF.PRS

‘Mala said that John knows *her*.’

c. Distinction at C-I, identity at SM

mala₁ [mala₂-r vailin₁-ta] b^heje p^helet^he
Mala Mala-GEN violin-CLF break.NF drop.PRF.PRS

‘Mala₁ has broken [Mala₂’s violin].’

d. Distinction at C-I, distinction at SM

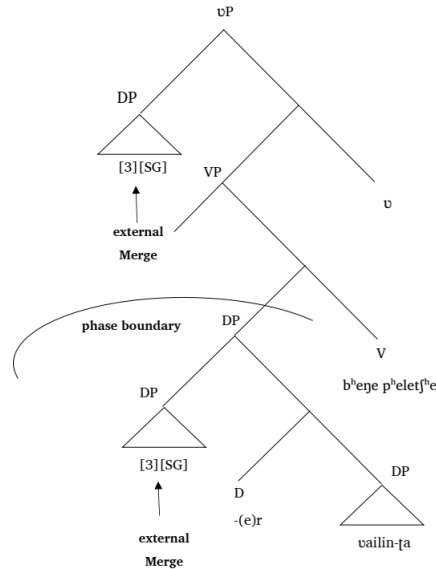
i. *mala₁ o₂-ke t^hene*
Mala 3SGL2-ACC know.INDF.PRS
‘Mala₁ knows her₂.’

ii. *mala₁ bollo dze jon o₂-ke t^hene*
Mala say.PFV C John 3SGL2-ACC know.INDF.PRS
‘Mala₁ said that John knows her₂.’

iii. *mala₁ [o₂-r vailin₁-ta] b^heje p^helet^he*
Mala 3SGL2-GEN violin-CLF break.NF drop.PRF.PRS
‘Mala₁ has broken [her₂ violin].’

The derivation of (20a) and (20c) and (20d-iii) is in Figure 8. Note that they all have the same syntactic derivation. They will receive distinct interpretations when a discourse rule applies such that it forces the reading 1 = 2 in the case of (20a). The morphosemantic feature [REF], along with the abstract root and morphophonological information, will determine whether *Mala* is realised or *or* is.

Figure 8: Derivation of (i) *mala mala-r vailin-ta b^{he}je p^{he}let^{he}e*, where the two *Mala*-s refer to the same individual in the given world, (ii) *mala mala-r vailin-ta b^{he}je p^{he}let^{he}e*, where the two *Mala*-s refer to distinct individuals in the given world, and (iii) *mala or vailin-ta b^{he}je p^{he}let^{he}e*, where *or* is free/deictic



5.2.3 LC-3 — leads to neither the creation of copies nor the creation of repetitions

C-I demands that the two linguistic items be interpreted as referring to distinct individuals. SM demands that the two linguistic items look distinct.

- (21) Neither Copies nor Repetitions
Distinction at C-I, distinction at SM

maloti mala-ke b^halobaʃe
Maloti Mala-ACC love.INDF.PRS.3L2

‘Maloti₁ loves Mala₂.’

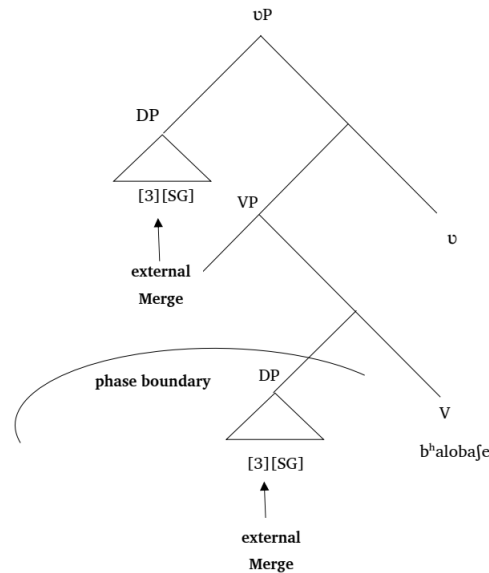
Finally, the case ‘neither copies not repetitions’ i.e., (21), is provided in Figure 9. Here too morphosemantic features along with the root and morphophonological information will lead to one DP being realised as *Maloti* and the other as *Mala*.

5.2.4 The Late Insertion Rule

- Internal Merge is able to copy morphosyntactic features. It cannot copy morphophonological features or morphological features, meaning abstract roots of lexical items cannot be copied.
- So how does the morphophonology know, in case of the derivation of, say, *John likes his house*, to translate the lower copy into a reflexive and the upper copy into the R-expression.
- I make a proposal here that has two parts.
 - The first part says that the lower copy of morphosyntactic features is accompanied by the morphosemantic feature [NREF]. [NREF], being a purely morphosemantic feature is inaccessible to the syntax. It is not copied and is only visible to the C-I interface and to the morphology.
 - The second part of the proposal assumes that all lexical items — including himself — undergo late insertion.
- A nominal is realised as an R-expression in the absence of the [NREF] feature. This is stated in the form of a morphological principle.

- (22) The principle of reference resolution

Figure 9: Derivation of *maloti mala-ke b^halobafe*, where *Maloti* and *Mala* refer to distinct individuals in the given world



- If [NREF] present, insert SELF-form or Pron-form
- Elsewhere, insert R-expression

6 Conclusion

- Being the simplest operation and having backing that it is innate from acquisition studies, Merge is a better candidate than Agree to drive the syntax of reference resolution. Along with argument manipulation (including reflexivization) that happens in the syntax subject to phasal restrictions, Merge can account for most cases of reference resolution.
- However, one has to test if this system can be extended to explain the binding of empty categories.

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